H01 runtime parent

Paired FASTQ streams

Shared input mutex

30 MB total chunk per worker

About 180 chunks 1-2 per active worker

12.93 s active-worker finish spread

A02 accepted in blackstar.2

Same FASTQ streams

Measured opportunity

Mapping threads at least 64?

Yes

No

1 MB total chunk bounded by read reserve

Retain configured legacy granularity

A02 changes scheduling granularity; serialized parsing still remains

Same locked parser same record boundaries same mapping kernels

Qualified public evidence at 96 threads

Uncompressed full corpus 70.88 s to 61.47 s 13.41% median gain

Gzip full corpus 71.68 s to 65.37 s 8.74% median gain

Median peak RSS down 6.77%

6 of 6 paired output checks plus canonical BAM pass